the US) video. A 640x480 image with 24 bits of data representation for colour and a frame rate of 30 per second easily gulps down a staggering 27 MB and that's without sound. Streaming or downloading such a video would be next to impossible. This is where compression comes into action.

The digital media currently available such as VCDs do not project good quality images due to variations and drawbacks in the existing digitisation methods. Moreover, these movies require greater space and hence you would need more than two to three CDs for a single movie. The current digital video standard used by video coders in SVCD (Super VCD) is MPEG-2. SVCD is midway between DVD and VCDs. MPEG-2 does not keep full copies of every frame of a video; it preserves the details of only those frames where there is a change. DVD does compensate with higher precision, preserving frames for perfect picture delivery and better sound effects. But the cost of a DVD is higher and the amount of space occupied by a DVD movie would be more (3.5 GB at a maximum). Thus, existing standards have certain drawbacks, which lead to a need for an improved technology, providing better picture quality, high-quality sound and a drastic reduction in the size of the movie.

DLP projectors from Texas Instruments

It's live!

Streaming video has been around for quite some time, and the video technologies till date range from low-quality streaming audio with an occasional picture to traditional download-and-play systems for a variety of file formats. The issues impeding the popularity of streaming video are compression and bandwidth. About the latter, nothing can be done unless technology improves and low-cost, high-bandwidth services are obtainable in the market. Compression is something that has seen improvement down the years. Microsoft, Apple and RealNetworks are some of the key players in the Internet streaming video arena, and streaming of video over the Net is currently usually done with standard formats developed by these companies.

Streaming works on the principle of buffer-and-deliver method. When the user clicks on a video, the associated application is launched and the video is buffered packet-by-packet on the client side and relay starts once enough packets have been acquired from the server. Thus, the user need not wait for a full download of the clip that he wants to view. Streaming has the advantage of live relay and also store-and-relay service over the Internet. But the issue here is one of quality—of picture and sound.

With newer Codecs, images would be sharper and near-DVD quality. The method and channel of distributing video and movies will change if these technologies prove to be successful in the market.

Already Apple is working on Darwin, a streaming media server that will allow you to start your own Internet broadcasting company for news, entertainment and educational programmes. It's built on the industry-standard Internet protocols, RTP (Real-Time Protocol) and RTSP (Real-Time Streaming Protocol). This enables the video-on-demand service by serving clients with QuickTime movies of their choice. Companies like Cisco and Sun are also taking part actively in the streaming video market. Texas Instruments is working on DLP or Digital Light Processing. It claims that DLP Cinema has the ability to project clearer, more realistic digital images than the current generation of analog movie projectors.

It's important to realise that the development of these technologies will mean much more than just developing a set of improved Codecs.

Honey, I shrunk the films

DivX is the new kid on the block, changing the way video will be delivered in the near future. It's a highly efficient video compression file format adapted from Microsoft's MPEG-4 Codec, removing some of its limitations, such as its inability to save files in AVI format. With DivX-based encoding program, a full-length movie in letterbox or full-screen format can be encoded with near-DVD quality and saved onto a single 650 MB CD! The compression ratio is an astounding 44:1 from AVI to DivX format.

This makes it the next step in compression technology for video delivery on the Internet. DivX uses the MP3 Codec for sound and hence the sound quality of DivX movies is enhanced.

Open Sesame

At present, ripping a DVD is illegal, so is putting

Compressing Data

A Codec, used for compressing video files, is any technology that can compress and decompress data. Codecs can be implemented in software, hardware, or a combination of both. An example of a hardware Codec would be the DVD decoder cards that accompany many DVD drives in place of such software as Power DVD or WinDVD. Codecs can simply be looked upon as plug-ins to a system to allow interpretation of a type of file that the operating system cannot recognise. For instance, if you download a movie trailer that has been encoded in MPEG4 format using the DivX Codec, you will need the same Codec installed to watch the trailer. For more information on Codecs and their functionalities visit **www.codecs.com**

Most compression methods are either Lossy or Lossless. Lossy compression doesn't perfectly represent what was compressed, but it's close enough given the size savings. Lossless compression guarantees that what goes in at one end comes out at the other end and hence compression in this case causes no loss in quality. Lossless compression isn't really a consideration when dealing with video over the Internet. Compression always involves a trade-off between picture/sound qualities and file size, with larger files obviously resulting in longer download times.